PAM Detector Module 054 Test Report and Datasheet

Built and tested March 2026 by Jay Frothingham and Colin Rush

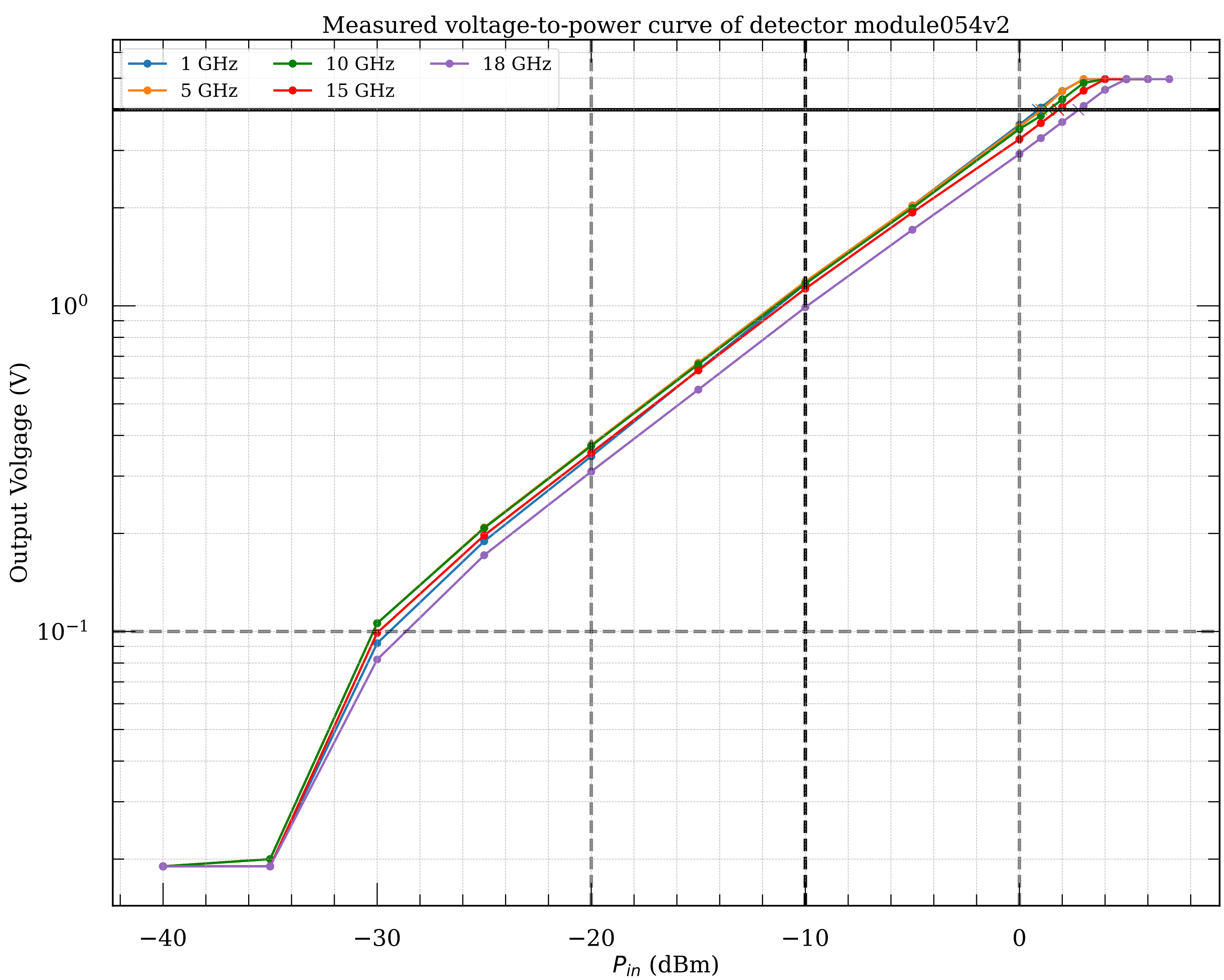
Notes

First build failed due to impedance mismatch at low frequencies. Success after rework.

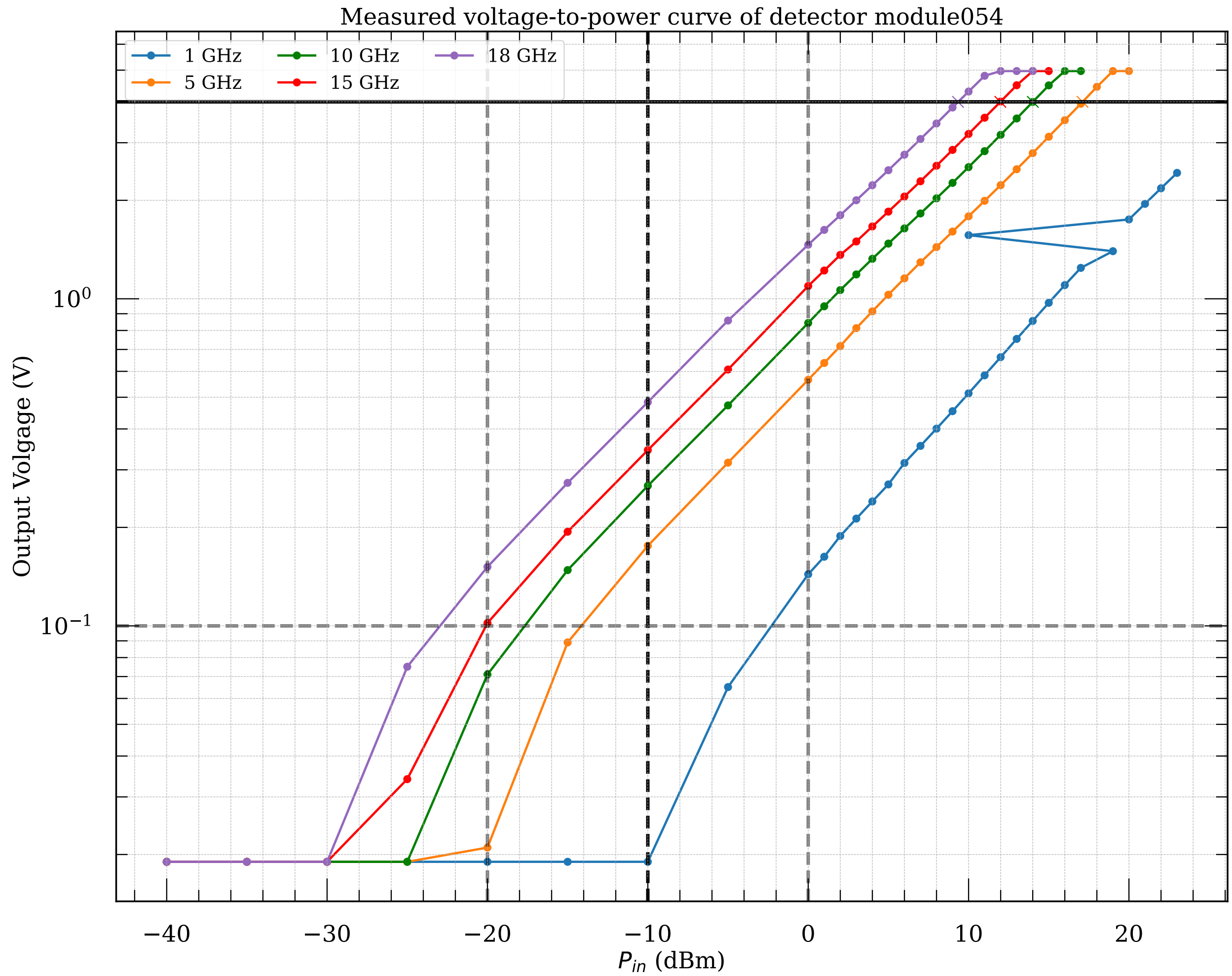
Voltage-to-power Curve

A signal generator (E4420B) was used to apply an RF input with a fixed frequency and varying power levels. It was connected with a 14 inch long SMA cable (AFX-CA-141-14 AtlanTec RF) to the detector module. The power was increased stepwise and the corresponding output voltages were measured using a multimeter. A power supply provided the +5 V supply voltage for the detector module.

Reworked module P-V curve



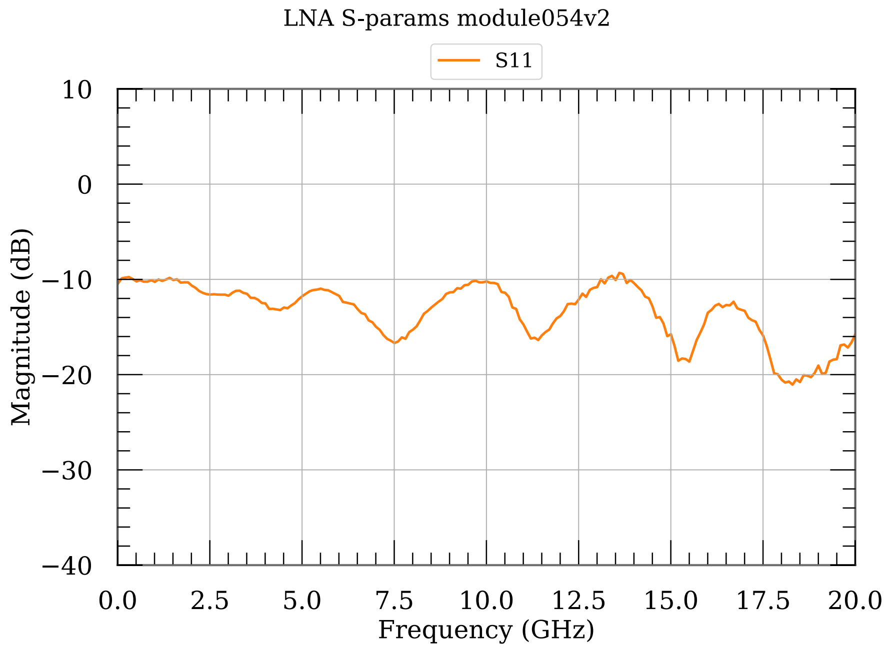
First build P-V curve



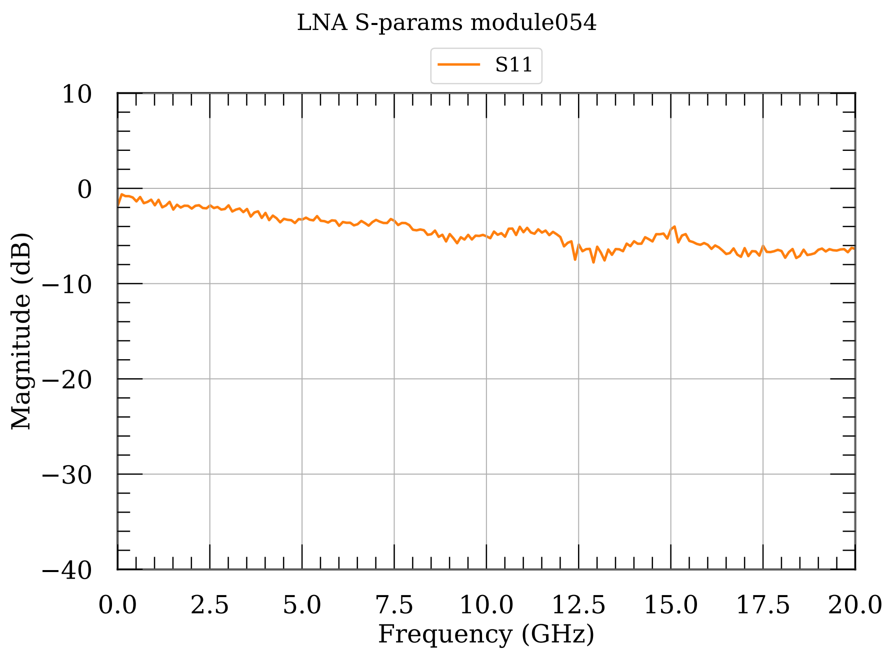
S-parameter

S-parameter measurement with the VNA (N5230C) of detector module.

Reworked module S-parameters

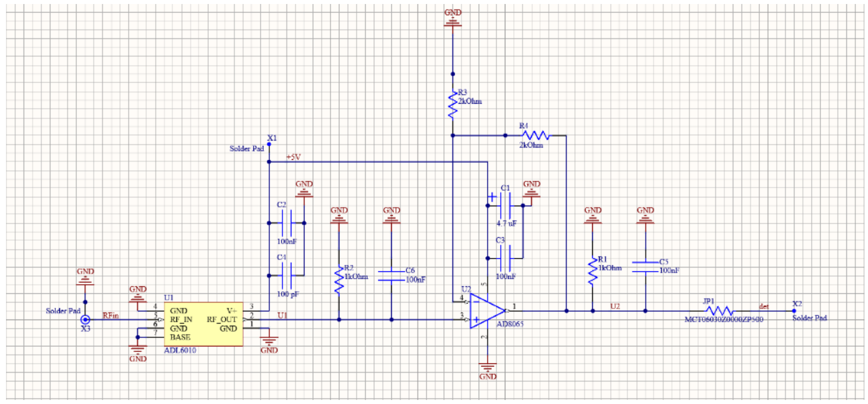


First build S-parameters



Schematic

- Frequency: 0.5-20GHz
- Impedance: 50Ω



Inputs:

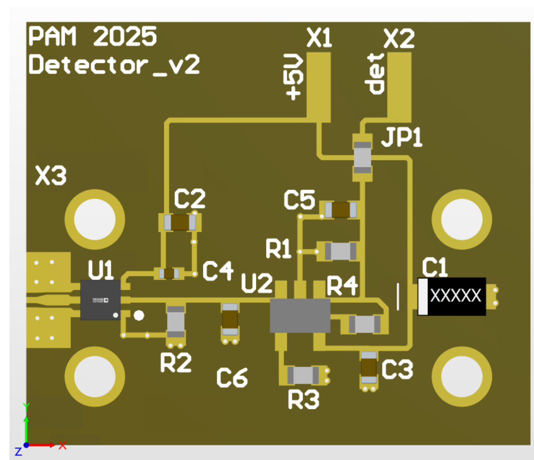
- RFin: Radio signal input
- +5V: positive supply voltage (+5V@1.6mA)
- GND

Outputs:

- Det: Output for power measurement

Components, Footprint - *All Case Codes are metric*

1 – SMA 2Hole (142-1701-201) [X3]
2 – Feed through cap: 4-40UNC-2A (B3C153B) [X1,X2]
1 – Turret Terminal 2-56 UNC – 2A (1595-2)
1 – detector LFCSP-6 (ADL6010ACPZN-R2) [U1]
1 – op amp SOT-23-5 (AD8065ARTZ-REEL7) [U2]
1 – 100pF 1005 (GRM1555C2A101FA01D) [C4]
1 – 4.7uF 3216 (TH3A475K020C5000) [C1]
4 – 100nF 1608 (GCM188R71C104KA37J) [C2-3,C5-6]
1 – 0Ω 1608 (MCT06030Z0000ZP500) [JP1]
1 – 1MΩ 1608 (MCT06030C1004FPW) [R1]
1 – 2kΩ 1608 (MCT06030C2001FP500) [R2]
1 – 1.6kΩ 1608 (MCT06030C1601FP500) [R4]
1 – 402Ω 1608 (MCT06030C4020FP500) [R3]



1 – RO4350 PCB
1 – Box
1 – Lid
2 – Screws 3-48 UNC - 2B x 3/16 (92196A091)
4 – Screws 2-56 UNC - 2B x 1/8 (21202)
4 – Screws 2-56 UNC - 2B x 5/32 (91771A884)
1 – RF-absorber PSA 0.08'', ca. 20 x 24 mm (MR42-0008-20)